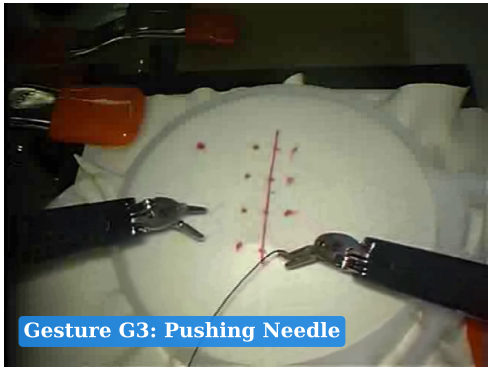
JIGSAWS Surgical Activity Video Benchmark: Multi-Task Environments & Intra-Trial Gesture Progression

Task 1: Suturing (39 Trials)
Trial B001, Frame 800 (t=26.7s)



Task 2: Knot Tying (36 Trials)
Trial B001, Frame 220 (t=7.3s)



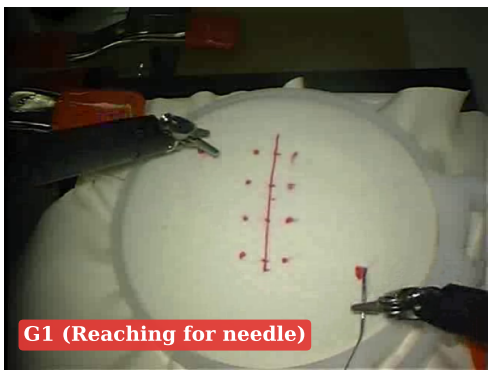
Task 3: Needle Passing (28 Trials)
Trial B001, Frame 1020 (t=34.0s)



JIGSAWS Visual Sensing Profile

- **Camera :** Stereo da Vinci Endoscope
- **Resolution :** 640 × 480 @ 30 FPS
- **Channels :** Left/Right RGB video
- **Environment :** Inanimate phantom
- **Subjects :** 8 surgeons (B--I)
- **Total Scale :** 103 trials (~285k frames)
- **Visual Encoders :** SurgMotion (1024D) and ResNet-18 (512D) frozen features

Stage 1: Reaching
Frame 150 (t = 5.0s)



Stage 2: Orientation
Frame 450 (t = 15.0s)



Stage 3: Penetration
Frame 800 (t = 26.7s)



Stage 4: Suture Pull
Frame 2400 (t = 80.0s)

